MVP TECH DOCUMENT

3.1 Introduction to MVP Approach

The Minimum Viable Product (MVP) for LevelFit AI is designed with the objective of **validating the core concept of gamified fitness engagement** while minimizing development complexity and resource usage.

Instead of building a fully AI-powered system from the beginning, the MVP focuses on delivering the **core engagement loop**, which includes:

- Task completion
- Reward allocation
- Progress tracking
- Accountability enforcement

This approach ensures that:

- Development time is reduced
- Core assumptions can be tested early
- User feedback can be collected efficiently

3.2 Core MVP Objective

The primary objective of the MVP is to answer the following key question:

👉 *"Can gamification improve user retention and consistency in fitness applications?"*

To achieve this, the MVP prioritizes:

- User engagement over feature richness
- Simplicity over complexity
- Performance over scalability (initial phase)

3.3 Core Functional Modules

1. User Authentication Module

This module ensures secure access to the application.

Features:

- Email and password-based login
- OAuth-based login (Google)
- Session management

Technical Considerations:

- Password encryption using hashing algorithms
- Token-based authentication (JWT/session tokens)
- Secure API communication

2. Onboarding System

The onboarding system is critical for personalization.

Steps involved:

1. Goal selection
2. Body metrics input
3. Medical conditions
4. Activity level
5. AI-based recommendation
6. Level assignment

Importance:

- Defines user journey
- Improves engagement from Day 1
- Enables personalized experience

3. Level-Based Progression Engine

This is the core of the gamification system.

Structure:

- 21 levels
- Each level = 5 days
- Total = 105 days

Logic:

- User completes daily tasks
- Points are accumulated
- Level unlocks after threshold

Design Justification:

- Short levels → quick wins
- Long journey → long-term engagement

4. Workout Tracking Module

This module allows users to log physical activity.

Features:

- Daily workout list
- Mark exercises as complete
- Input sets and reps
- Track duration

System Behavior:

- On completion → triggers scoring engine
- Updates user progress instantly

5. Diet Logging Module

Tracks user nutrition intake.

Features:

- Meal categorization (breakfast, lunch, dinner, snacks)
- Calorie calculation
- Macro tracking (protein, carbs, fats)

Importance:

- Ensures holistic fitness tracking
- Integrates with scoring system

6. Scoring Engine Module

The scoring engine processes all user actions.

Key Responsibilities:

- Assign points
- Maintain streak
- Trigger level progression
- Handle reset logic

Why Important:
👉 It drives engagement, motivation, and habit formation

7. Progress Dashboard

Provides visual feedback to users.

Displays:

- Current level
- Total points
- Streak count
- Daily progress

Impact:

- Reinforces motivation
- Makes progress visible

8. Achievement System

Provides milestone-based rewards.

Examples:

- Streak badges
- Level completion badges

Purpose:

- Psychological reward system
- Enhances user satisfaction

3.4 Technology Stack Justification

Frontend (React.js)

- Component-based architecture
- High performance
- Easy scalability

Styling (Tailwind CSS)

- Fast UI development
- Responsive design

Backend (Appwrite / Node.js)

- Serverless architecture
- Easy API handling
- Scalable

Database (PostgreSQL)

- Structured data handling
- Reliable and secure

Hosting (Vercel)

- Fast deployment
- CDN-based performance

3.5 Limitations of MVP

To keep the system simple, certain features are excluded:

- AI-based food recognition
- Smart workout generation
- Wearable integration
- Computer vision tracking

These are planned for future phases.

3.6 Expected Outcomes of MVP

- Validate user engagement model
- Measure retention rates
- Analyze user behavior
- Identify improvement areas